

REAPER-Notator

Complete User Manual & Functional Reference

Native Score Editing, Engraving & DAW Automation for Cockos REAPER

Version:	v1.0.1 (Production Release)
Release Date:	October 03, 2026
Compatibility:	Cockos REAPER v7.0+ (Windows, macOS, Linux) with RealmGui
License:	GNU General Public License v3.0 (GPL-3.0)

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Chapter 1: Introduction & Interface Architecture

1.1 Overview & Vision

REAPER-Notator is a native, professional music scoring, notation editing, and engraving environment built directly into Cockos REAPER using RealmGui and standard SMuFL (Standard Music Font Layout) Bravura vector glyphs. Unlike external scoring applications, Notator runs directly inside REAPER's audio engine and timeline, providing instantaneous two-way synchronization between visual music notation and underlying REAPER MIDI takes, CC automation, and tempo maps.

1.2 The Three-Column Workspace Layout

The user interface is designed around an ergonomic three-column layout:

* Left
augme

1.3 Window Resizing & Docking

REAPER-Notator can run as a floating window or docked directly into any REAPER docker (bottom, top, left, right, or multi-tabbed). The score canvas dynamically scales according to the zoom factor (adjustable from 0.6x to 2.0x), preserving crisp vector typography at all DPI scaling factors.

Chapter 2: Top Toolbar & Global Navigation

2.1 Title & Version Indicator

Located in the upper left corner, displaying 'REAPER Notator' alongside the active build and semantic version number (e.g. v1.0.1). When running in a developer environment, Notator automatically detects and displays the active Git commit hash.

2.2 Transport & Playhead Controls

Functions:

* Rew

2.3 Timeline & Meter Displays

Functions:

* Mea

2.4 Audition Preview (Note Audio on Click)

Function: When enabled (speaker icon), clicking or vertically dragging any note on the canvas triggers an immediate acoustic preview sound.

DAW E
Audio I

2.5 Follow Playhead (Auto-Scroll)

Function: When active, the score canvas scrolls horizontally in real time to keep the active playback cursor centered in the viewport.

2.6 View Mode Selector

Functions:

* Auto

2.7 Track Picker & Visibility Filters

Function: Opens a multi-track routing popover allowing the user to select which REAPER tracks are currently visible, focused, or edited on the score canvas.

Chapter 3: Score Canvas & Note Editing Engine

3.1 Note Entry & Pitch Snapping

Functions:

* Click

3.2 Note Duration & Augmentation Dots

Functions:

* Note

3.3 Accidentals (Sharps, Flats, Naturals)

Functions:

* Flat

3.4 Note Manipulation (Move, Pitch, Length)

Functions:

* Hori

3.5 Multi-Note Selection & Marquee Tool

Functions:

* Ma
measu

3.6 Second-Interval Collision & Voice Separation

Functions:

* Se
accord

3.7 Standard Gould Rests

Functions:

* Emp

Chapter 4: Standard-Compliant Automatic Beaming Engine

4.1 Meter-Based Beat Grouping

Functions: Consecutive eighth, 16th, and 32nd notes automatically connect with solid beams. Beaming boundaries strictly observe meter division (e.g. 4/4 groups into two half-measure halves; 6/8 groups into two dotted-quarter pulses of three eighths each).

4.2 Slope Calculation & Melodic Contour

Functions: Beams calculate a graceful visual slant following the pitch direction of noteheads. Extreme melodic intervals apply Gould slant-clamping to prevent excessive angles, and flat passages render strictly horizontal beams.

4.3 Fractional Beams (Beamlets & Stubs)

Functions: Syncopated rhythms and mixed subdivisions (such as a dotted eighth followed by a 16th note) generate standard fractional beamlets (stubs) oriented toward the rhythmic pulse beat.

4.4 Stem Inversion ('X' Shortcut)

Functions: Pressing the 'X' key instantly flips the stem direction of selected notes (stems up vs. stems down). Inverting stems automatically recalculates beam anchor points and flag alignments.

Chapter 5: Dynamics Engine, Hairpins & Bow Swell Function

5.1 Standard Dynamic Levels (ppp to fff)

Functions: The Dynamics Drawer features 10 one-click dynamic badges:

* ppp

5.2 Multi-Target CC Automation

Functions: Dynamics can write continuous automation curves to:

* CC1

5.3 Hairpins (Crescendo & Diminuendo)

Functions:

* Cres

5.4 Bow Swell Function & Bow Position Slider

Functions:

* Bow
back d

5.5 Phrasing & Shaping Bypass

Function: Checkbox to temporarily disable CC automation playback without deleting visual dynamic markings on the score canvas.

Chapter 6: Articulations & Rearticulate Integration

6.1 Rearticulate Bank Auto-Discovery

Functions: REAPER-Notator automatically scans REAPER's user directory for Rearticulate sound bank definitions (Rearticulate.reabank). When a track is selected, Notator matches track names (e.g. 'Violin I', 'Cello', 'Horns') to corresponding library banks.

6.2 Visual Playing Technique Glyphs

Functions: Provides an instant-access drawer for score playing techniques:

* Stac

6.3 DAW Automation Impact

DAW Effect: Inserting an articulation writes:

1. RE

Chapter 7: Holding & Sustain Pedal Lane (CC64)

7.1 Dedicated Pedal Lane

Functions: Renders continuous piano sustain pedal markings positioned below the bass staff with automatic collision clearance from low notes and dynamic hairpins.

7.2 Three Historical Engraving Styles

Functions:

* Clas

7.3 Dual Handles & Pause / Break Retakes

Functions:

* Star

Chapter 8: Tempo Maps & Metric Transitions

8.1 Absolute Tempo Markers

Functions: Insert tempo markers with custom BPM values (e.g. Quarter = 120, Dotted Quarter = 72) and descriptive Italian tempo text (Adagio, Andante, Allegro, Presto).

DAW E

8.2 Gradual Transitions (Accelerando / Ritardando)

Functions: Renders dashed tempo transition lines spanning multiple measures (e.g. 'poco a poco accel. -----').

DAW E

Chapter 9: Octave Shift Lines (8va, 8vb, 15ma, 15mb)

9.1 Visual Score Simplification

Functions: For extreme high or low passages that would otherwise require excessive ledger lines, octave shift lines simplify score reading:

* 8va

9.2 Visual vs. Sounding Pitch

DAW Effect: Notes remain in their natural visual position on the staff for effortless reading, while underlying MIDI note pitches trigger the intended high or low acoustic octaves.

Chapter 10: Orchestral Pattern Browser

10.1 Built-in Cinematic Pattern Library

Functions: Slide-out pattern browser organized into 8 orchestral categories:

1. Str

10.2 One-Click Insertion & Capture

Functions:

* Inse

Chapter 11: Text Items & Score Annotations

11.1 Rehearsal Marks & Structural Tags

Functions: Pressing Ctrl+T inserts a floating score text item at the cursor. Supports rehearsal letters ([A], [B], [C]), section titles ('Verse', 'Chorus', 'Bridge'), and orchestration performance instructions ('Molto espressivo', 'Solo', 'Tutti').

11.2 In-Place Editing & Typography

Functions: Double-click any text item to open an in-place editing field. Configure font size, standard/italic/bold styling, and staff attachment anchor points.

Chapter 12: MusicXML 4.0 Import & Export

12.1 Interoperability Standard

Functions: MusicXML 4.0 is the universal interchange format between professional notation software. REAPER-Notator includes a dedicated MusicXML parser and exporter written in pure Lua.

12.2 Importing Scores

Functions: Click 'Import MusicXML' to load orchestral scores created in external notation programs. Notator parses parts, measures, time signatures, key signatures, pitch data, dynamics, and tempo markers, creating fully arranged tracks and MIDI items directly in REAPER.

12.3 Exporting Scores

Functions: Click 'Export MusicXML' to save your REAPER project as a standard `.musicxml` file ready for publication, live orchestral recording sessions, or further engraving.

Chapter 13: Page Print & PDF Score Layout

13.1 Print Modal & Layout Setup

Functions: Dedicated Print Settings dialog accessible from the top bar tools:

* Pap

13.2 Metadata & Publishing Header

Functions: Inscribe Title, Subtitle, Composer, Arranger, and Copyright notices in classical engraving typography.

Chapter 14: Keyboard Shortcuts & Quick Reference

14.1 Comprehensive Hotkey Matrix

Quick reference table for high-speed score entry:

Key / Shortcut	Action	Description
:---	:---	:---
Space	Play / Pause	Starts or stops timeline playback
Delete / Backspace	Delete	Deletes selected notes, dynamics, tempo marks, or lines
Esc	Clear Selection	Deselects all active items and clears focus
Up / Down Arrows	Transpose Pitch	Moves selected notes up/down by 1 semitone
Shift + Up / Down	Octave Shift	Moves selected notes up/down by 1 octave (12 semitones)
Left / Right Arrows	Nudge Position	Shifts notes horizontally by the active grid increment
Shift + Left / Right	Resize Duration	Shortens or lengthens selected notes by grid step
X	Invert Stem	Flips note stem direction (up <-> down)
1 .. 6	Rhythmic Values	1=Whole, 2=Half, 3=Quarter, 4=Eighth, 5=16th, 6=32nd
. (Period)	Toggle Dot	Toggles dotted note duration (1.5x)
- / 0 / =	Accidentals	-=Flat (b), 0=Natural (nat), ==Sharp (#)
T	Tie Note	Toggles tie / Bindebogen to adjacent note
S	Slur Phrase	Toggles legato slur over selected passage
A	Accent	Toggles accent mark (>) on selected note
Ctrl + T	New Text Item	Creates a floating text annotation at cursor position
Ctrl + S	Save Project	Triggers REAPER project save